

COLLABORATIVE DISCUSSION 2: SUMMARY POST

Ariel Mella

12701097

Unit 10

Machine Learning

University of Essex Online

The highlight was discovering that the posts broadly coincide in seeing rapid progress in LLM capabilities and deployments alongside tensions around rights, transparency and ethics.

My initial post focused on LLMs' evolution and their productivity gains (Noy and Zhang, 2023), for example through models such as GPT-5.5, while also providing the illusion of "simulated cognition" (Hutson, 2021; Shojaee et al., 2025). This gap, coupled with the rapid adoption of LLMs through agentic tools and chatbots, surfaced concerns and made me question whether mainstream use can produce a form of cognitive outsourcing that may affect human critical thinking in the long term (Sparrow, Liu and Wegner, 2011).

Peer feedback enriched this perspective by grounding these risks in structural and field-specific context. Bakri reinforced the threat of cognitive dependency, linking it to the "Google effect" (Sparrow, Liu and Wegner, 2011): why would we remember information if we can rely on Google to retrieve it for us? Conversely, Bulat challenged the causal link of my analogy to social media, proposing a deployment framework instead.

Furthermore, the forum exchange surfaced a paradox in the writing and creative domains: while LLMs can augment individual abilities, they simultaneously reduce diversity by producing more similar outputs (Doshi and Hauser, 2024).

Forum posts broadly identified the advantages of LLMs through references and use cases within the writing and related fields, while at the same time areas such as ethics (Floridi et al., 2018), copyright (United States Copyright Office, 2025), risks and transparency (Resnik and Hosseini, 2025) surfaced as points to be addressed. Reflecting on the Industry 5.0 readings, I can see the parallelism with the shift from a techno-economic to a socio-technical future, and how the forum discussion suggests that LLMs are not simply productivity tools, but socio-technical systems whose individual benefits may generate collective risks (Morales, Nousala and Ghobakhloo, 2024; Nousala, Metcalf and Ing, 2024).

To conclude, from the evidence I have seen in these short discussions and the specific readings, there is evidence to support that LLMs deliver benefits in many areas related to writing and creativity; however, their technical progress and adoption are advancing faster than governance can adapt to address risks associated with it.

References

- Doshi, A.R. and Hauser, O.P. (2024) 'Generative AI enhances individual creativity but reduces the collective diversity of novel content', *Science Advances*, 10(28), article eadn5290. Available at: <https://doi.org/10.1126/sciadv.adn5290>.
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C., Madelin, R., Pagallo, U., Rossi, F., Schafer, B., Valcke, P. and Vayena, E. (2018) 'AI4People—an ethical framework for a good AI society: opportunities, risks, principles, and recommendations', *Minds and Machines*, 28(4), pp. 689–707. Available at: <https://doi.org/10.1007/s11023-018-9482-5>.
- Hutson, M. (2021) 'Robo-writers: the rise and risks of language-generating AI', *Nature*, 591(7848), pp. 22–25. Available at: <https://doi.org/10.1038/d41586-021-00530-0>.
- Morales, M., Nousala, S. and Ghobakhloo, M. (2024) 'The complexity of sustainable innovation, transitional impacts of Industry 4.0 to 5.0 for our societies: circular society exploring the systemic nexus of socioeconomic transitions', in Nousala, S., Metcalf, G. and Ing, D. (eds) *Industry 4.0 to Industry 5.0: explorations in the transition from a techno-economic to a socio-technical future*. Singapore: Springer Nature Singapore, pp. 31–56. Available at: <https://doi.org/10.1007/978-981-99-9730->

- Nousala, S., Metcalf, G. and Ing, D. (eds) (2024) *Industry 4.0 to Industry 5.0: explorations in the transition from a techno-economic to a socio-technical future*. Singapore: Springer Nature Singapore. Available at: <https://doi.org/10.1007/978-981-99-9730-5>.
- Noy, S. and Zhang, W. (2023) 'Experimental evidence on the productivity effects of generative artificial intelligence', *Science*, 381(6654), pp. 187–192. Available at: <https://doi.org/10.1126/science.adh2586>.
- Resnik, D.B. and Hosseini, M. (2025) 'Disclosing artificial intelligence use in scientific research and publication: when should disclosure be mandatory, optional, or unnecessary?', *Accountability in Research*, pp. 1–13. Available at: <https://doi.org/10.1080/08989621.2025.2481949>.
- Shojaee, P., Mirzadeh, I., Alizadeh, K., Horton, M., Bengio, S. and Farajtabar, M. (2025) 'The illusion of thinking: understanding the strengths and limitations of reasoning models via the lens of problem complexity', *arXiv*, 2506.06941 [Preprint]. Available at: <https://doi.org/10.48550/arXiv.2506.06941>.
- Sparrow, B., Liu, J. and Wegner, D.M. (2011) 'Google effects on memory: cognitive consequences of having information at our fingertips', *Science*, 333(6043), pp. 776–778. Available at: <https://doi.org/10.1126/science.1207745>.
- United States Copyright Office (2025) *Copyright and artificial intelligence, part 2: copyrightability*. Washington, DC: United States Copyright Office. Available at: <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-2-Copyrightability-Report.pdf> (Accessed: 23 June 2026).